5 A progressive wave is produced at $x = 0$ on the surface of a liquid. As the wave travels, its amplitude gradually decreases. At one particular time, the variation of the vertical displacement y with distance x along the surface of the liquid is shown in Fig. 5.1.

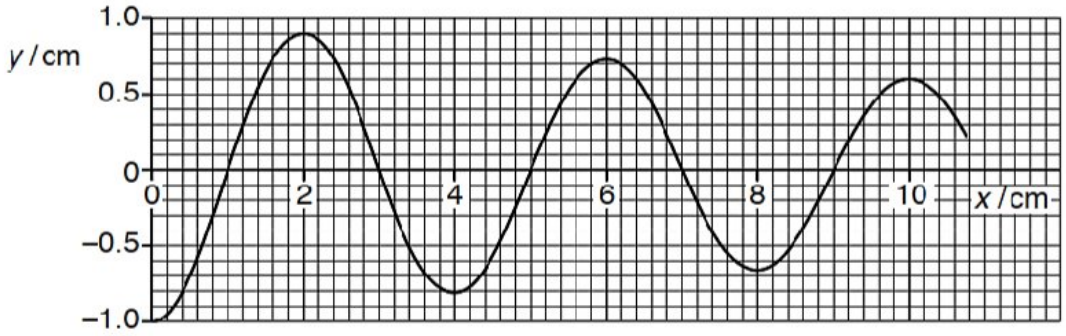


Fig. 5.1

(a) State what is meant by a *progressive wave*.

.....

.....

..... [1]

(b) From Fig. 5.1, determine the wavelength of the wave

wavelength = ... m [1]

(c) The frequency of the wave is 12.5 Hz. Hence determine the speed of the wave.

speed = ... $m\ s^{-1}$ [1]

(d) A particle Q is at $x = 4.4$ cm. Mark and label on Fig. 5.1. the position of a particle P that lags Q in phase by 5.03 rad. Show your workings clearly. [2]

(e) The amplitudes of the first and third crests are shown in Fig. 5.1.

Determine the fraction of the intensity at the third crest to that at the first crest.

fraction = ... [2]